Remmogo PDF Report API Contract

Laboratory 8 service contract for administrative PDF report generation

1 Purpose and traceability

This contract defines the operation used by a Remmogo Administrator to generate a printable PDF report of rental transactions. It supports **FR-13 Account Management and Reporting**, the **Generate reports** administrator use case, **BR-08 Administrative Reporting**, and **QS-05 Administrative Report Generation**. Under QS-05, a semester report containing more than 500 records must be ready within 10 seconds.

The PDF rendering application is an external service. Remmogo remains the owner of account, listing and rental data. The external service receives only the prepared report content needed to render the document and does not query the Remmogo database directly.

2 Operation summary

Item	Contract
Remmogo endpoint	POST /api/v1/admin/report-exports
Authentication	Bearer access token
Authorization	ADMINISTRATOR role required
Request content type	application/json
Success content type	application/json
Idempotency	Required Idempotency-Key header
Downstream dependency	PDF Rendering App POST /v1/documents
Expected outcome	A stored REPORT_EXPORT record and a downloadable PDF reference

3 Remmogo request

Headers

```
Authorization: Bearer <access-token>
Content-Type: application/json
Idempotency-Key: 3f90a462-9d1e-4e6b-bf7b-4bb4c861a4f1
```

Body

```
{
  "reportType": "SEMESTER_RENTALS",
  "periodStart": "2026-08-01",
  "periodEnd": "2026-12-20",
  "statusFilter": ["COMPLETED", "ACTIVE", "OVERDUE"],
  "includeUserDetails": true
}
```

Field validation

Field	Required	Validation

Field	Required	Validation
reportType	Yes	Must be a supported value. Initial value: SEMESTER_RENTALS.
periodStart	Yes	ISO 8601 date. Must not be after periodEnd.
periodEnd	Yes	ISO 8601 date. Must not be before periodStart or in an unapproved future reporting period.
statusFilter	No	Each value must be a valid RENTAL_STATUS; an omitted or empty list means all statuses.
includeUserDetails	No	Boolean; defaults to false. Detailed data remains restricted to Administrators.
Idempotency-Key	Yes	UUID or opaque value of 16-80 characters. The same key and body return the same export; the same key with a different body is rejected.

4 Processing rules

1. Authenticate the caller and verify the `ADMINISTRATOR` role.
2. Validate the request before creating an export record.
3. Reuse an existing result when the idempotency key and request fingerprint match.
4. Create `REPORT_EXPORT` with status `PROCESSING` in a transaction.
5. Query a consistent snapshot of qualifying rentals. The PDF app never receives database credentials.
6. Send the prepared title, period, generation timestamp, column definitions and report rows to the PDF Rendering App.
7. If rendering succeeds, store the returned file reference and checksum, change the status to `COMPLETED`, and return the result.
8. If the dependency times out or is unavailable, change the status to `FAILED` with a stable failure code. A retry with the same idempotency key must not create a second export.

5 Downstream PDF Rendering App contract

Request

```
POST /v1/documents
Content-Type: application/json
X-Correlation-ID: 6d46b4c4-2e37-4bfb-a8bc-4b53af91afd8
{
  "documentType": "table-report",
  "outputFormat": "PDF",
  "title": "Remmogo Semester Rental Report",
  "fileName": "remmogo-rentals-2026-s1.pdf",
  "metadata": {
    "reportId": "8c964f13-0f07-420c-b45d-98aeaa086610",
    "periodStart": "2026-08-01",
    "periodEnd": "2026-12-20",
    "generatedAt": "2026-10-12T17:00:00Z"
  },
  "columns": [
    {"key": "rentalId", "label": "Rental ID"},
    {"key": "itemName", "label": "Item"},
    {"key": "customerName", "label": "Customer"},
    {"key": "ownerName", "label": "Owner"},
    {"key": "status", "label": "Status"},
    {"key": "expectedReturnDate", "label": "Due Date"}
  ],
  "rows": [
    {
```

```
      "rentalId": "R-1042",
      "itemName": "Mini fridge",
      "customerName": "Example Customer",
      "ownerName": "Example Owner",
      "status": "ACTIVE",
      "expectedReturnDate": "2026-10-20"
    }
  ]
}
```

Successful rendering response

```
HTTP/1.1 201 Created
Content-Type: application/json
{
  "documentId": "pdf_01JAB6QTTQSCJGK3",
  "status": "COMPLETED",
  "downloadUrl": "https://pdf-app.example/documents/pdf_01JAB6QTTQSCJGK3",
  "sha256": "d37c5f1c4f84c9c8f0b6b928c882c8d8074bde6a58f56c07166bf832431f3d3e",
  "expiresAt": "2026-10-13T17:00:00Z"
}
```

6 Remmogo success response

```
HTTP/1.1 201 Created
Location: /api/v1/admin/report-exports/8c964f13-0f07-420c-b45d-98aeaa086610
Content-Type: application/json
{
  "exportId": "8c964f13-0f07-420c-b45d-98aeaa086610",
  "status": "COMPLETED",
  "reportType": "SEMESTER_RENTALS",
  "periodStart": "2026-08-01",
  "periodEnd": "2026-12-20",
  "recordCount": 527,
  "downloadUrl": "/api/v1/admin/report-exports/8c964f13-0f07-420c-b45d-98aeaa086610/file",
  "checksum": "d37c5f1c4f84c9c8f0b6b928c882c8d8074bde6a58f56c07166bf832431f3d3e",
  "completedAt": "2026-10-12T17:00:06Z"
}
```

Remmogo exposes its own authenticated download route rather than returning the external service URL to the browser.

7 Stable error outcomes

All errors use this structure:

```
{
  "error": {
    "code": "PDF_SERVICE_TIMEOUT",
    "message": "The report could not be generated at this time.",
    "correlationId": "6d46b4c4-2e37-4bfb-a8bc-4b53af91afd8",
    "retryable": true
  }
}
```

HTTP	Stable code	Meaning	Client action
400	INVALID_REPORT_REQUEST	Dates, status values or required fields are invalid.	Correct the request; do not retry unchanged.
401	AUTHENTICATION_REQUIRED	Token is absent, expired or invalid.	Sign in again.

HTTP	Stable code	Meaning	Client action
403	ADMIN_ROLE_REQUIRED	Authenticated caller is not an Administrator.	Do not retry with the same account.
409	IDEMPOTENCY_KEY_REUSED	The key was previously used with a different body.	Submit with a new key.
422	NO_REPORT_DATA	No rentals match the approved filters.	Change the filter or period.
503	PDF_SERVICE_UNAVAILABLE	Rendering dependency is unavailable.	Retry with the same idempotency key.
504	PDF_SERVICE_TIMEOUT	Rendering did not complete within the configured timeout.	Retry with the same idempotency key.
500	REPORT_EXPORT_FAILED	An unexpected internal failure occurred.	Contact support with the correlation ID.

8 Security and privacy

- Only Administrators may generate or download administrative reports.
- Remmogo sends only fields required for the selected report.
- Access tokens and password hashes are never included in rendering payloads.
- External download URLs are not exposed directly to clients.
- Transport uses TLS, service credentials are stored outside source code, and logs exclude report-row personal data.
- Generated files have an expiry/retention policy and an audit record identifies the requesting Administrator.

9 Failure and recovery behaviour

The principal failure risk is a timeout after the PDF app has accepted a request but before Remmogo receives the response. Without idempotency, retrying could generate duplicate files. Remmogo therefore stores the client idempotency key and a request fingerprint before calling the dependency, propagates a correlation identifier, and returns the existing export on a matching retry. A background reconciliation operation may query the PDF app by correlation identifier before starting another render.

Verification test: Simulate a timeout after the PDF app stores the document. Retry the identical request with the same idempotency key. Confirm that Remmogo returns one `REPORT_EXPORT`, one external document and the same `exportId`; no duplicate report is created.

10 Data impact

The operation reads `RENTAL`, `ITEM_LISTING`, `STUDENT` and related account data from a consistent snapshot. It writes one `REPORT_EXPORT` record. It does not modify rental status, item availability, payment confirmations or return confirmations.